Multivariate Angular Simulation

Edinburgh Summer School on Generative AI for Extremes

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Acknowledgements

This talk is based on the following paper:

Wessel, J. B., Murphy-Barltrop, C. J. R. and Simpson, E. S. (2026).
Generative machine learning for multivariate angular simulation.
Extremes, 29(2):195-243.

It was a contribution to the recent *Extremes* Special Issue on:

"Bridging heavy tails and artificial intelligence".

Huge credit and thanks go to:

- my co-authors, Callum and Jakob;
- Miguel and Jordan for their work organising the Special Issue.

Overview

1. Background: Angular-radial representations in multivariate extremes
2. Multivariate angular simulation: The role of machine learning
3. Diagnostic techniques
4. Simulation study and a metocean application
5. Summary and outlook

Motivation

Angular-radial representations are reasonably common in the study of multivariate extremes.

For a d-dimensional random vector $\mathbf{X} = (X_1, \ldots, X_d) \in \mathbb{R}^d$, it may be convenient to consider

$$R = \|\mathbf{X}\|_A, \qquad \mathbf{W} = \mathbf{X}/\|\mathbf{X}\|_B,$$

i.e., a *radial* component and a *(pseudo-)angular* vector.

Motivation - multivariate regular variation

Take $\mathbf{X}$ to have some common, heavy-tailed (e.g., Fréchet or Pareto) margins, and use the L_1 norm in both cases.

Then, $R > 0$ and $\mathbf{W}$ takes values on an angular simplex

$$\left\{ \mathbf{w} \in [0,1]^d : \sum_{i=1}^{d} w_i = 1 \right\}.$$

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Under the assumption of multivariate regular variation,

$$\lim_{t \to \infty} \Pr(R > tr, \mathbf{W} \in B \mid R > t) = \frac{H(B)}{r}.$$

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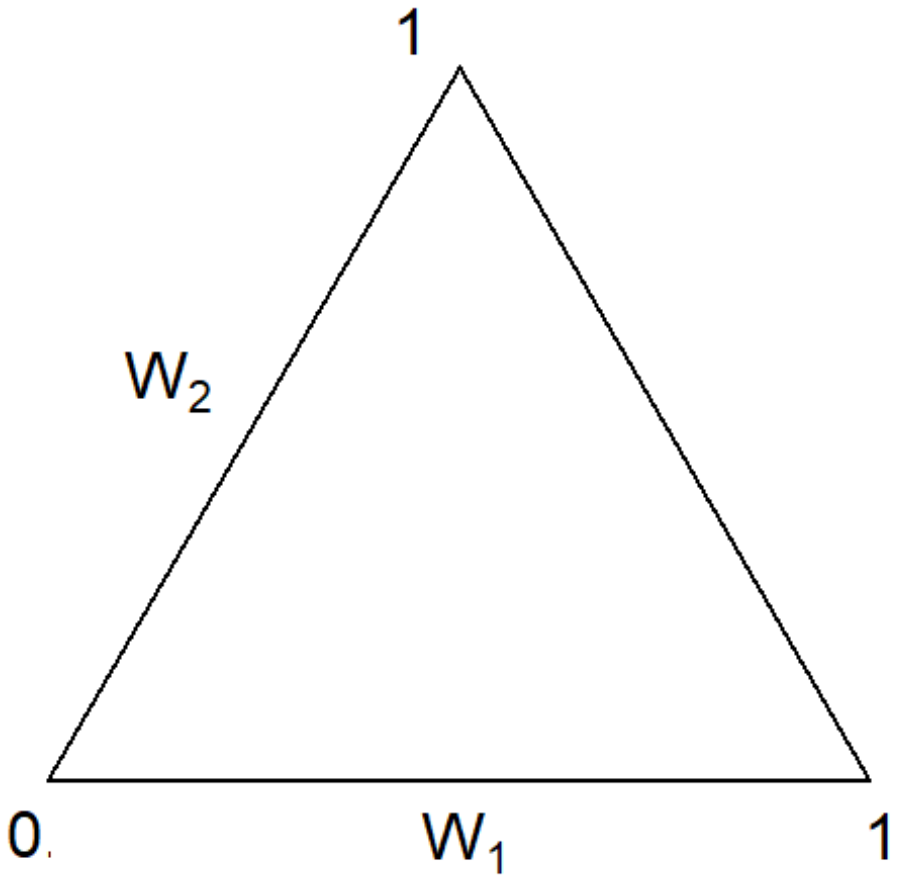
$$\lim_{t \to \infty} \Pr(R > tr, \mathbf{W} \in B \mid R > t) = \frac{H(B)}{r}.$$

- So, R and $\mathbf{W}$ become independent in the limit.
- The limiting spectral measure H gives information about the extremal dependence structure of $\mathbf{X}$ (e.g., Simpson et al. (2020)).

Mass on the angular simplex ($d = 3$)



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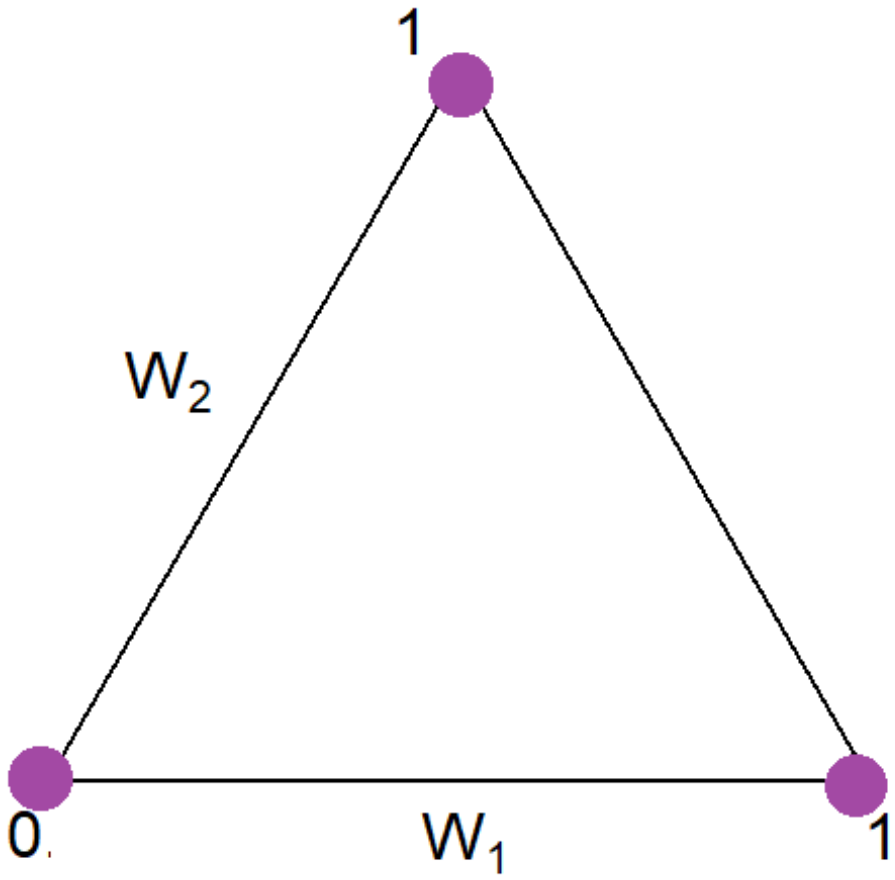

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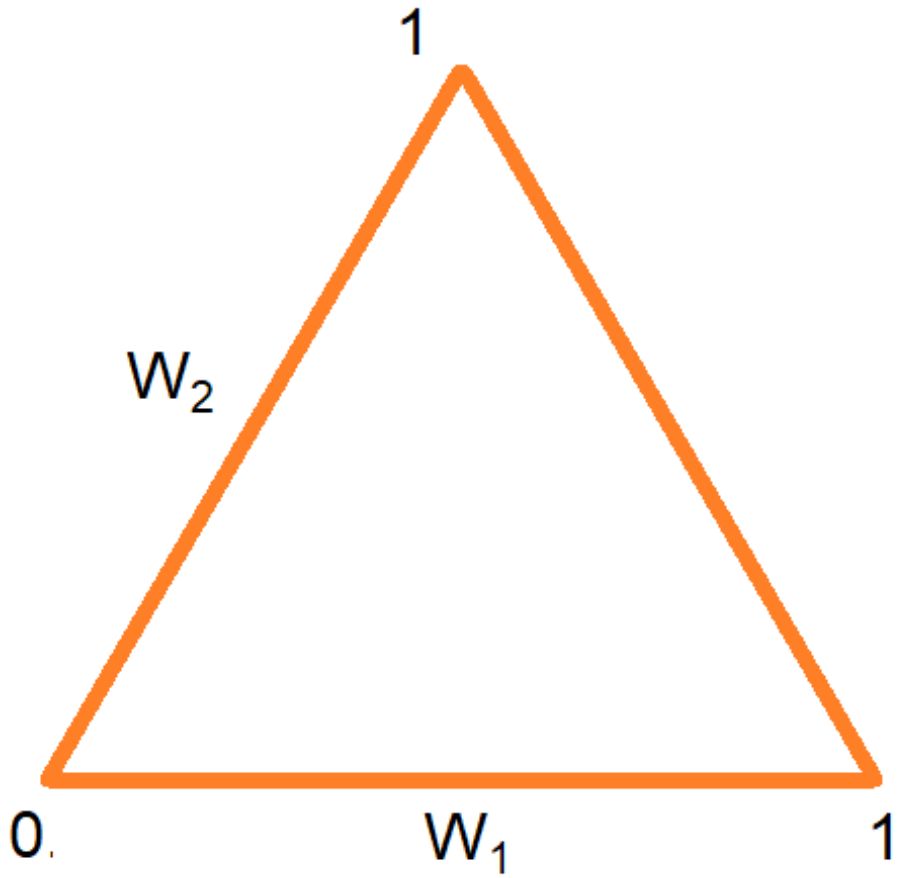

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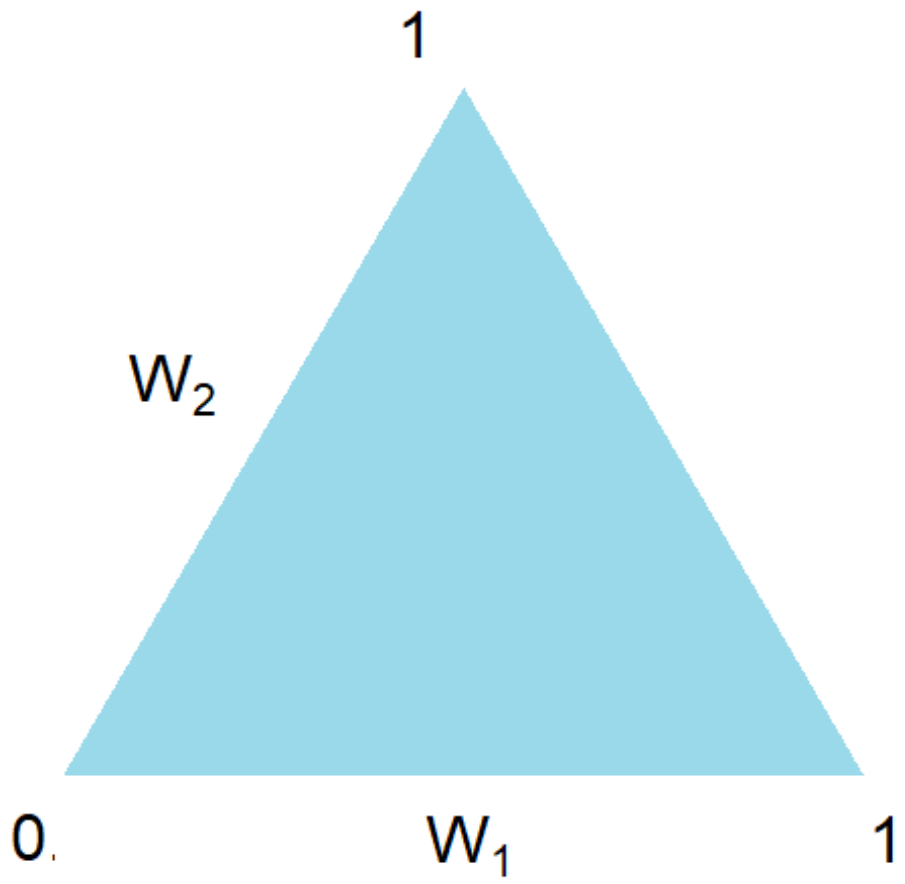

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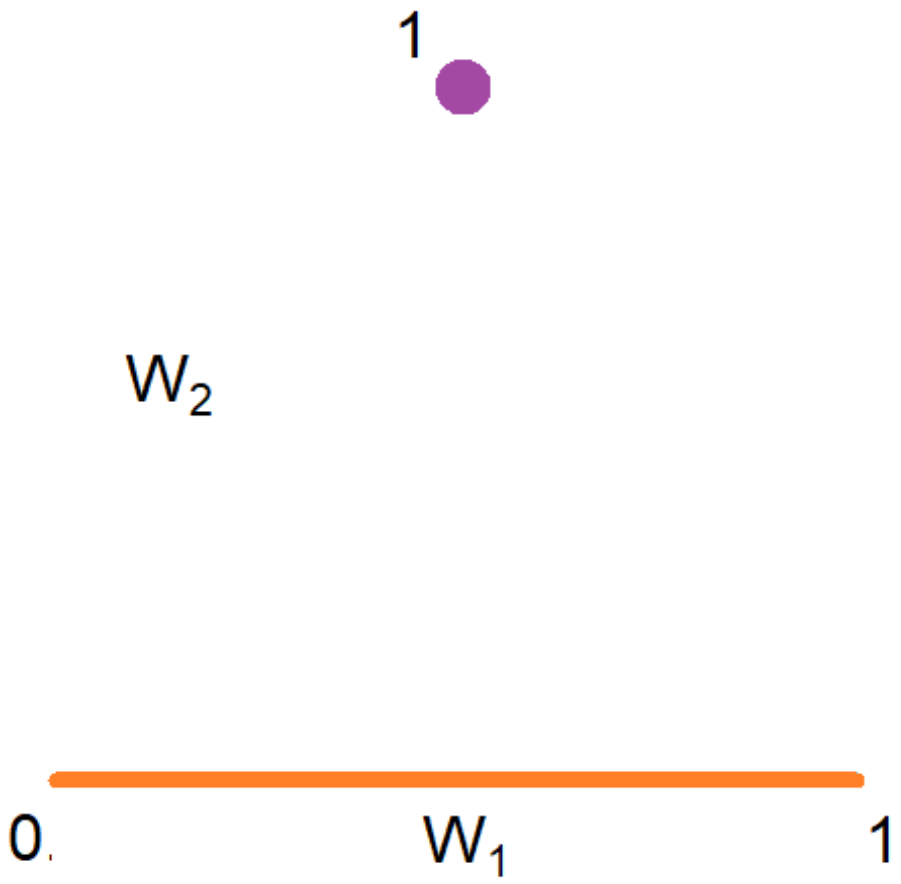

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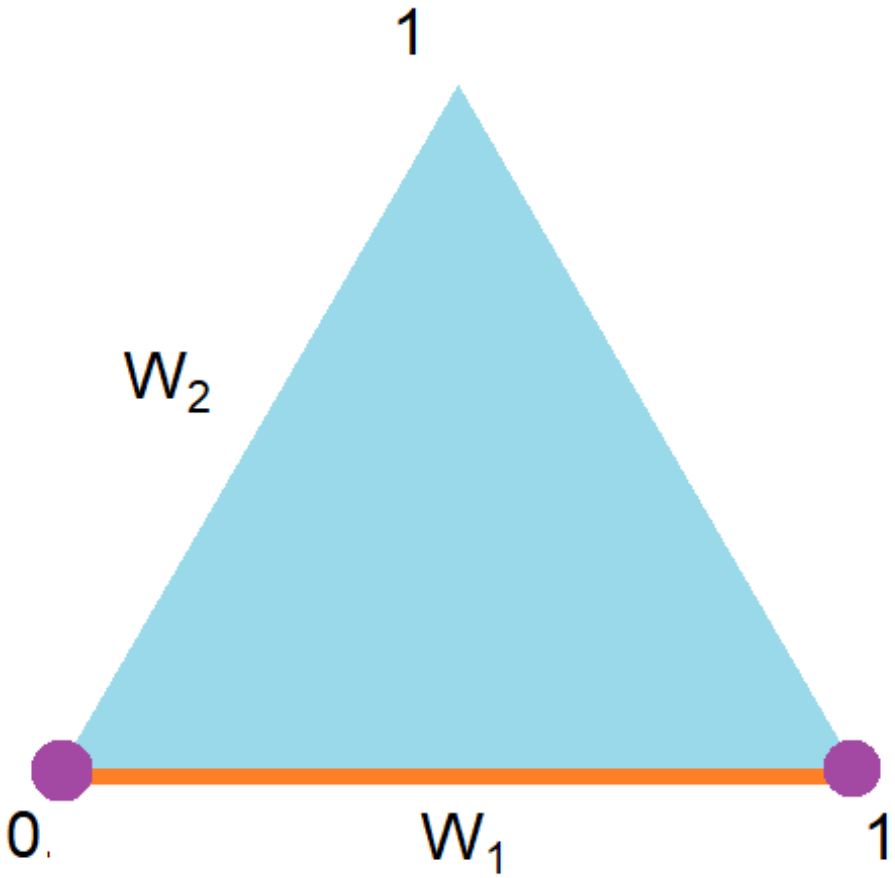

Motivation - radial-angular multivariate extremes

More recently, there has been growing interest in radial-angular approaches for multivariate extremes, e.g.:

- SPAR models;
- geometric extremes.

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- SPAR models;
- geometric extremes.

For inference and extrapolation, both require modelling or simulation of an angular component.

Motivation - SPAR models

- This idea originates with Mackay and Jonathan (2023),
  - Modelling multivariate extremes through angular-radial decomposition of the density function. *arXiv:2310.12711*.
- and there are several recent applied/inferential contributions, e.g.,
  - Mackay, Murphy-Barltrop and Jonathan (2025).
    The SPAR Model: A new paradigm for multivariate extremes: Application to joint distributions of metocean variables.
  - Murphy-Barltrop, Mackay and Jonathan (2025).
    Inference for bivariate extremes via a semi-parametric angular-radial model.
  - Mackay, Murphy-Barltrop, Richards and Jonathan (2026).
    Deep learning joint extremes of metocean variables using the SPAR model.

Motivation - SPAR models (main idea)

Consider a decomposition of the density of $(R, \mathbf{W})$ via

$$f_{R,\mathbf{W}}(r, \mathbf{w}) = f_{\mathbf{W}}(\mathbf{w})\, f_{R|\mathbf{W}}(r \mid \mathbf{w}).$$

Then, the multivariate extremes problem relies on estimation of an angular density $f_{\mathbf{W}}$ and a model for the tail of $R \mid \mathbf{W}$, i.e., for $r > u(\mathbf{w})$

$$f_{R,\mathbf{W}}(r, \mathbf{w}) = \tau f_{\mathbf{W}}(\mathbf{w}) f_{GPD}(r - u(\mathbf{w}); \sigma(\mathbf{w}), \xi(\mathbf{w})),$$

where $\tau = \Pr(R > u(\mathbf{w}) \mid \mathbf{W} = \mathbf{w})$ is close to 0.

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where $\tau = \Pr(R > u(\mathbf{w}) \mid \mathbf{W} = \mathbf{w})$ is close to 0.

- Extrapolation can be achieved by drawing from the angular distribution, then simulating from the conditional GPD.
- No requirement to assume fixed marginal distributions.

Motivation - Geometric extremes

- Suppose that $\mathbf{X}$ has light-tailed margins (e.g., exponential or Laplace).
- Consider the gauge function $g(\mathbf{x})$, defined via

$$-\log f(t\mathbf{x}) \sim t g(\mathbf{x}), \qquad t \to \infty.$$

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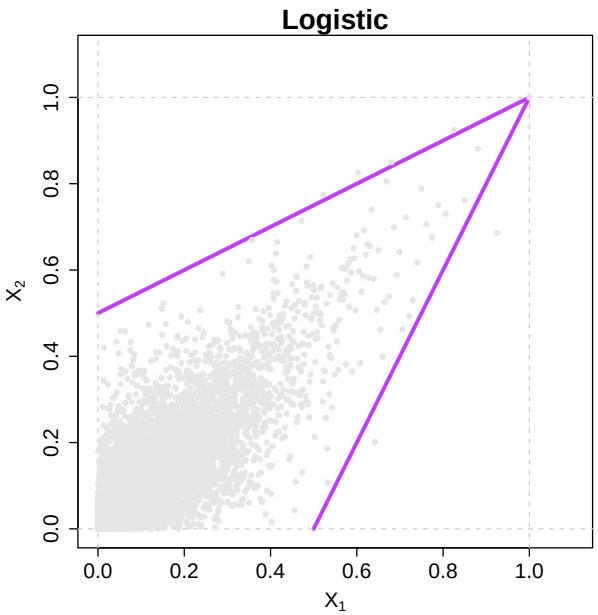
$$-\log f(t\mathbf{x}) \sim t g(\mathbf{x}), \qquad t \to \infty.$$

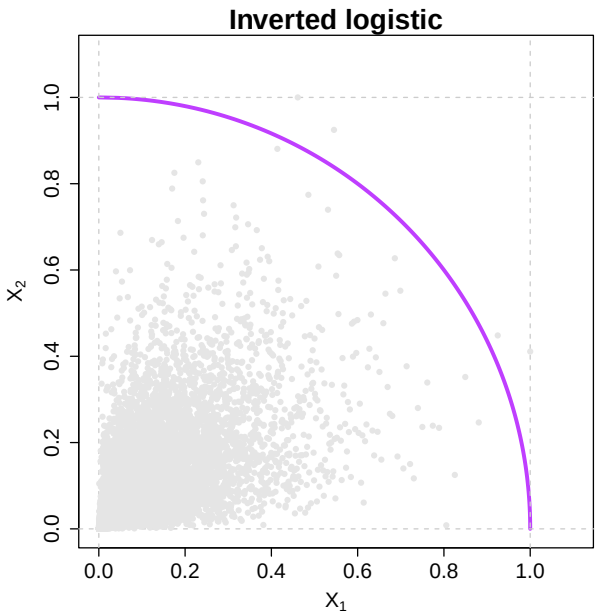
- Suitably scaled samples from $\mathbf{X}$ converge onto the set $\{\mathbf{x} : g(\mathbf{x}) \leq 1\}$, with boundary set

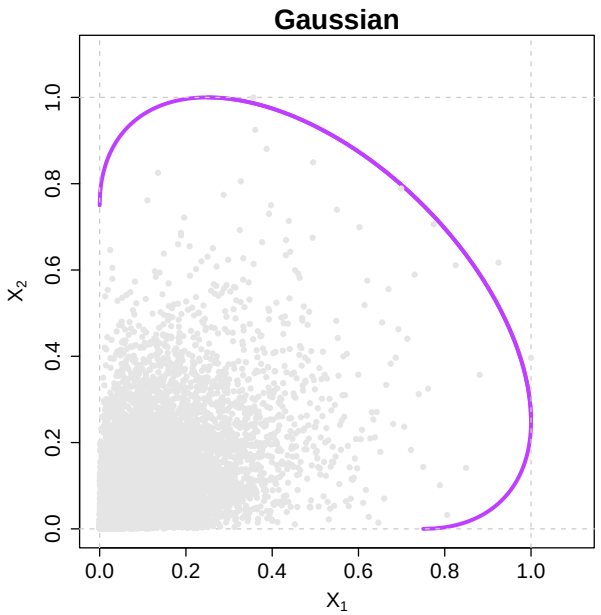
$$G = \{\mathbf{x} : g(\mathbf{x}) = 1\}.$$

- Nolde (2014) links G to the coefficient of tail dependence η.
- Nolde and Wadsworth (2022) make a connection to several different representations of multivariate extremes.

Motivation - Geometric extremes

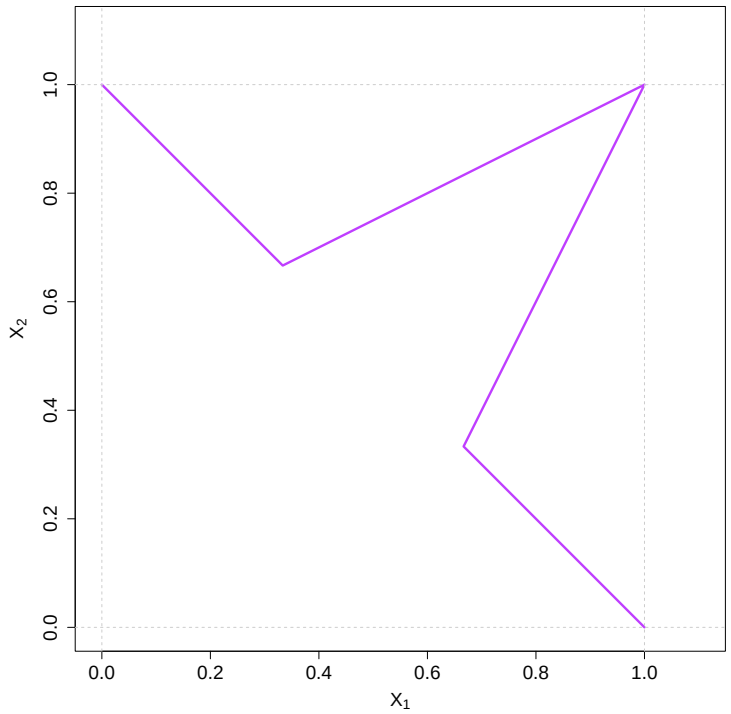






Motivation - Geometric extremes

- The limit set is star-shaped.
  - If it contains $\mathbf{x}$, it also contains $t\mathbf{x}$, for any $t \in (0, 1)$.



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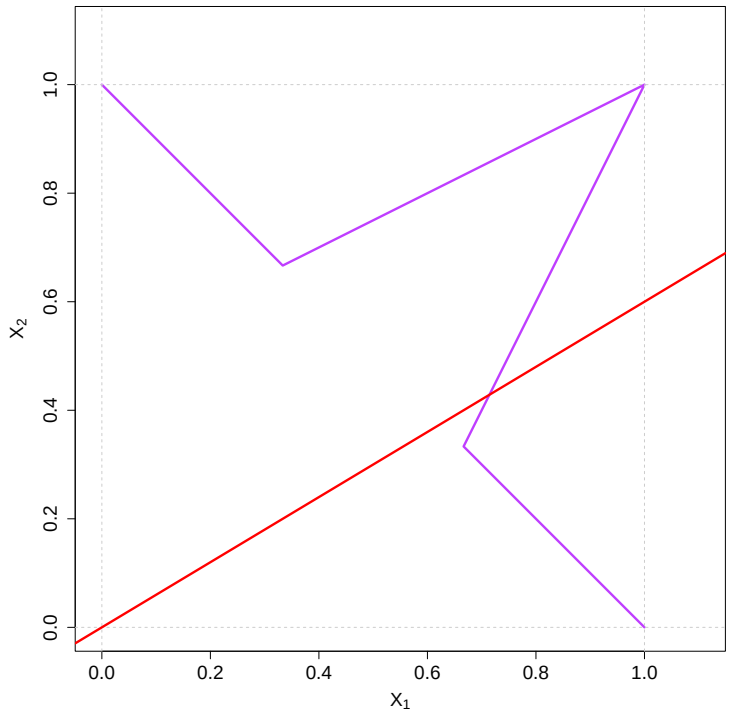

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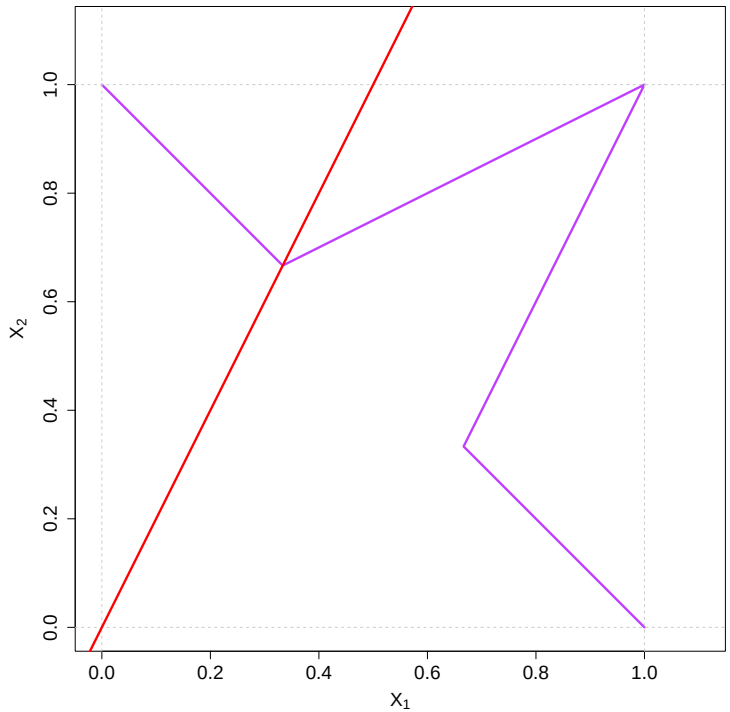

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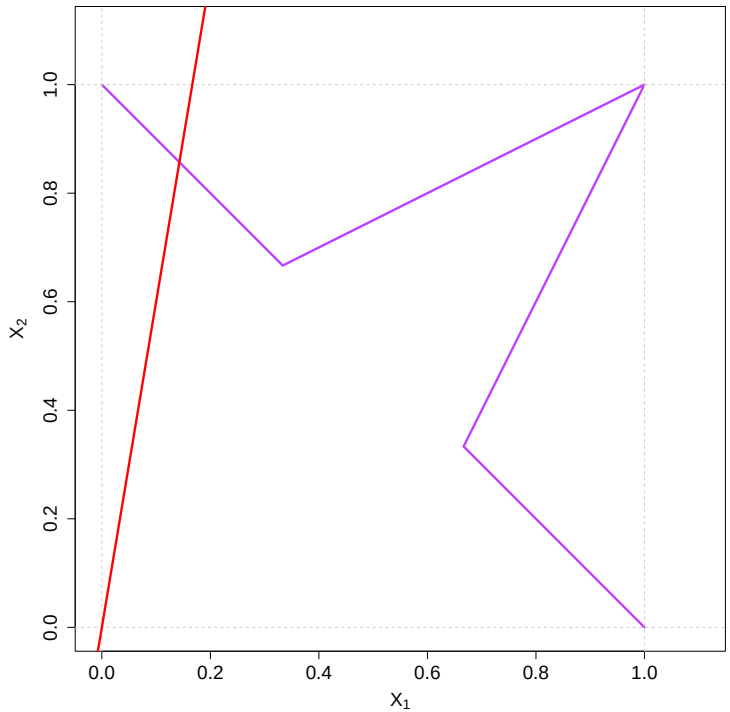

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- Wadsworth and Campbell (2024) exploit this framework for joint tail modelling, including estimation of joint tail probabilities.

Motivation - Summary

- Radial-angular representations are useful in the study of multivariate extremes.
- Different frameworks require different (or no) assumptions on the marginal distributions.
- A key step in these approaches is often to model (or simulate from) a multivariate angular distribution.

The role of AI in extreme value modelling

- Using AI approaches for extreme value analysis requires care.
- Standard approaches may not be well-suited to extrapolation tasks.

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- Using AI approaches for extreme value analysis requires care.
- Standard approaches may not be well-suited to extrapolation tasks.

- Options include:
  - using AI techniques for parameter estimation in theoretically-justified extreme value models;
  - adapting the AI and/or the extremes approach to suit the problem;
  - identifying sub-tasks within an extreme value analysis approach where AI may be able to help.

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Our focus - simulating angular variables

- Empirical approaches could be used to simulate from the angular distribution.
  - This becomes infeasible in higher dimensions.
- Parametric models could be used instead.
  - These may lack the flexibility required to cover the range of possible structures.

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- Parametric models could be used instead.
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- Simulation using generative deep learning approaches could be used.
  - This can estimate the angular distribution without specifying a density.
  - It is less restrictive than relying on exact observations.

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- So, modelling and simulating angular data can play an important role in multivariate extreme value analysis.

- This may also be useful in other settings where angular data arise.

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- So, modelling and simulating angular data can play an important role in multivariate extreme value analysis.
- This may also be useful in other settings where angular data arise.

Our question:
Can generative deep learning methods be helpful for this task?

- Note: We will focus on the L_2 (Euclidean) norm throughout.

Spherical coordinates

We focus on the spherical coordinates $\boldsymbol{\Theta} = (\Theta_1, \ldots, \Theta_{d-1})$

$$\Theta_1 = \text{atan2}\left(\sqrt{{X_d}^2 + {X_{d-1}}^2 + \cdots + {X_2}^2}, X_1\right),$$
$$\Theta_2 = \text{atan2}\left(\sqrt{{X_d}^2 + {X_{d-1}}^2 + \cdots + {X_3}^2}, X_2\right),$$
$$\vdots$$
$$\Theta_{d-2} = \text{atan2}\left(\sqrt{{X_d}^2 + {X_{d-1}}^2}, X_{d-2}\right),$$
$$\Theta_{d-1} = \text{atan2}\left(X_d, X_{d-1}\right),$$

where $\Theta_1, \ldots \Theta_{d-2} \in [0, \pi]$ and $\Theta_{d-1} \in (-\pi, \pi]$. (Blumenson, 1960)

Deep learning approaches

- Generative adversarial networks (GANs) - Goodfellow et al. (2014)
- Normalizing flows
  - Neural spline flows (NFNSFs) - Durkan et al. (2019)
  - Masked autoregressive flows (NFMAFs) - Papamakarios et al. (2017)
- Flow matching (FM) - Lipman et al. (2022)

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- Maximum mean discrepancy networks/Energy score networks

Adaptations for angular variables - GANs

- We can think of this as a 'competition' between a generator and a discriminator.
- We use feedforward neural networks for both components.
- To ensure the spherical coordinates lie in the correct range, we specify activation functions after the last generator layer as

$$\theta_i \leftarrow \pi \cdot \frac{1}{1 + e^{-x_i}} \text{ for } i = 1, \ldots, d-2,$$
$$\theta_{d-1} \leftarrow \pi \cdot \tanh x_{d-1}.$$

- Note: circular wrapping resulted in some training instabilities.

Adaptations for angular variables - NFs

- NFs involve a base distribution and a series of transformations.
- For both the NFNSF and NFMAF approaches, we use a transformation of the form

$$\theta_i \leftarrow \pi \cdot \Phi(x_i) \text{ for } i = 1, \ldots, d-2,$$
$$\theta_{d-1} \leftarrow (x_{d-1} \bmod 2\pi) - \pi.$$

- In these cases, the sigmoid has slower convergence at 0 and 1 than the Gaussian CDF, which led to poor results near the endpoints of $[0, \pi]$.
- Circular wrapping for the final component worked well in this case.

Adaptations for angular variables - FM

- For flow matching, we don't need to map onto the angular space, since flows can be defined directly on the sphere.

Baseline approach - mixture of vMF distributions

- The von Mises-Fisher (vMF) distribution has density

$$f_{vMF}(\mathbf{x} \mid \boldsymbol{\mu}, \kappa) = c_d(\kappa) e^{\kappa \boldsymbol{\mu}^T \mathbf{x}}, \quad \mathbf{x} \in \mathbb{S}^{d-1},$$

  for some concentration $\kappa \geq 0$, mean direction $\boldsymbol{\mu} \in \mathbb{S}^{d-1}$ and normalising constant $c_d(\kappa)$.

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- This is an angular extension of a multivariate Gaussian distribution.

- A mixture of vMF distributions provides a flexible, parametric approach to modelling angular data. This provides a baseline method for comparing our deep learning approaches.

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Evaluation strategies

We propose a range of evaluation metrics to assess both marginal and joint structure.

- Numerical metric:
  - Angular energy score (aES)
- Diagnostic plots:
  - Histograms and QQ-plots for spherical margins
  - Scatterplots between pairs of spherical angles
  - Orthant probability plots to assess full structure

Evaluation strategies: Angular energy score

- Scoring rules assess whether an estimated distribution represents observed data.
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- For $d > 2$, we propose a natural extension termed the *angular energy score* (aES).

- The aES is a proper scoring rule, but **not** strictly proper.

Evaluation strategies: Diagnostic plots

- Histograms and QQ-plots for spherical margins
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Evaluation strategies: Orthant probability plots

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Evaluation strategies: Orthant probability plots

- $\mathbb{S}^{d-1}$ can be divided into 2^d orthants.
- Let $p_k = \mathbb{P}(\mathbf{W} \text{ is in the } k\text{th orthant})$, for $k = 1, \ldots, 2^d$.
- We can compare empirical estimates of these quantities for observed and generated samples.
- Plotting on a log-scale is generally preferred.

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Simulation study - overview

To cover a wide range of examples, we consider the following options:

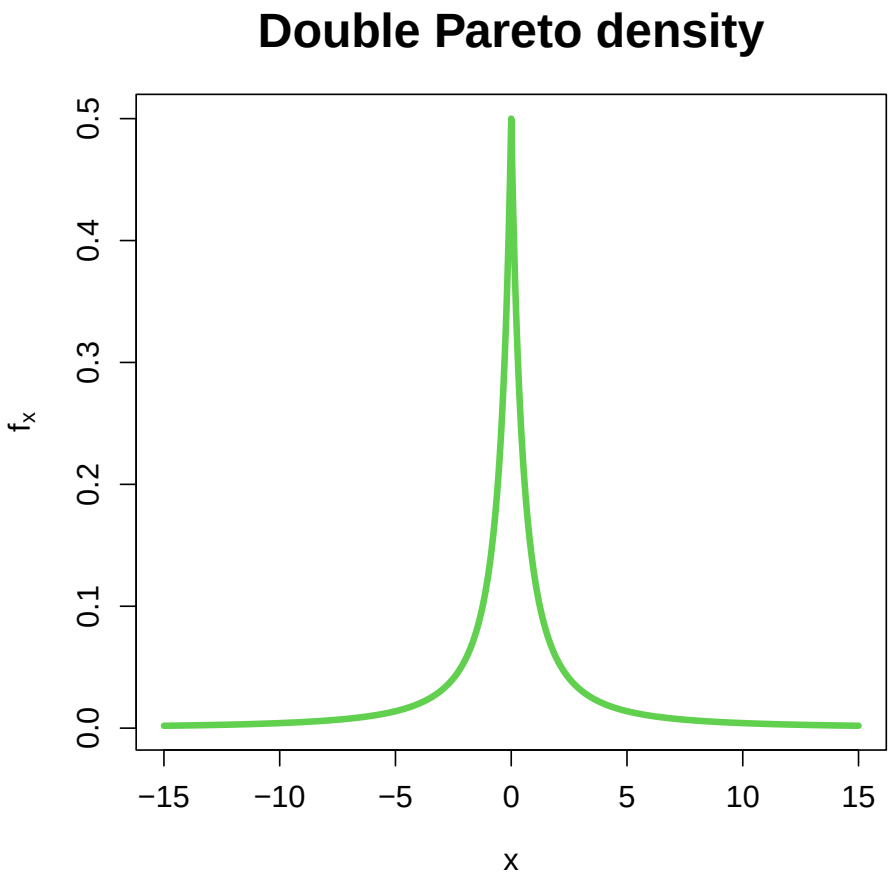
- dimensions $d = 5$ and $d = 10$;
- datasets of size $n = 1,000$, $n = 10,000$ and $n = 100,000$;
- light and heavy-tailed marginal distributions;
- five different dependence (copula) structures.

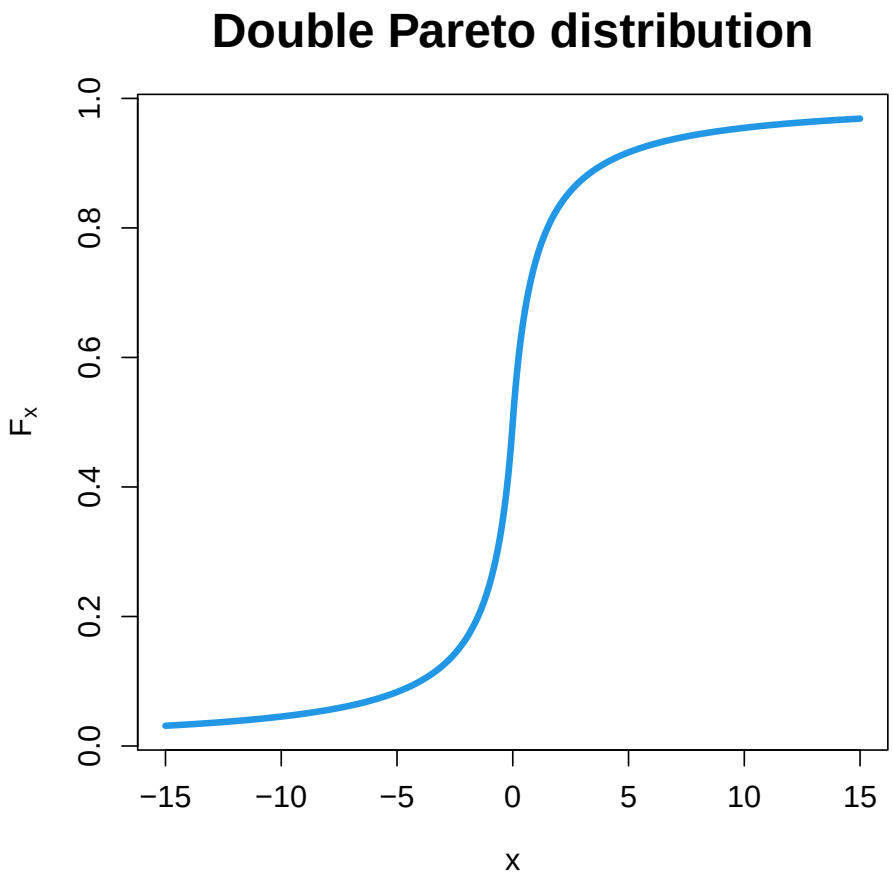
Simulation study - marginal models

To allow the potential for mass in all orthants, we choose common marginal models with support on $(-\infty, \infty)$.

- A standard Laplace distribution, having light upper and lower tails.
- A 'double Pareto' distribution, with heavy upper and lower tails.

Simulation study - marginal models





Simulation study - copula models

We consider copulas with different types of dependence structure, taking into account AD/AI and sparsity:

- a Gaussian copula (AI);
- a mixture of a Gaussian copula and a student-t copula (AI);
- a logistic copula (AD);
- a mixture distribution of a logistic copula and an independence copula (more complex EDS);
- a 'sparse' Gaussian copula specified such that the variables are clustered into groups of dependent variables with independence between groups (AI).

Simulation study - hyperparameter optimisation

- Bespoke hyperparameter tuning for each individual setting is very time consuming and therefore infeasible.
- We instead optimised the architectures for one specific case - the sparse Gaussian copula with double Pareto margins - and using a data example.

Simulation study - hyperparameter optimisation

- Bespoke hyperparameter tuning for each individual setting is very time consuming and therefore infeasible.
- We instead optimised the architectures for one specific case - the sparse Gaussian copula with double Pareto margins - and using a data example.
- This seemed reasonable as long as the hyperparameters that were selected allowed for some flexibility.
- Details of our suggested hyperparameters and architectures are in the paper.

Simulation study - general takeaways

- Using the aES, the baseline approach gave the best results in the majority of cases, but the results are very close.
  - Seems difficult to distinguish between methods using the aES.
  - Possibility that the aES focuses too much on the marginal fits.

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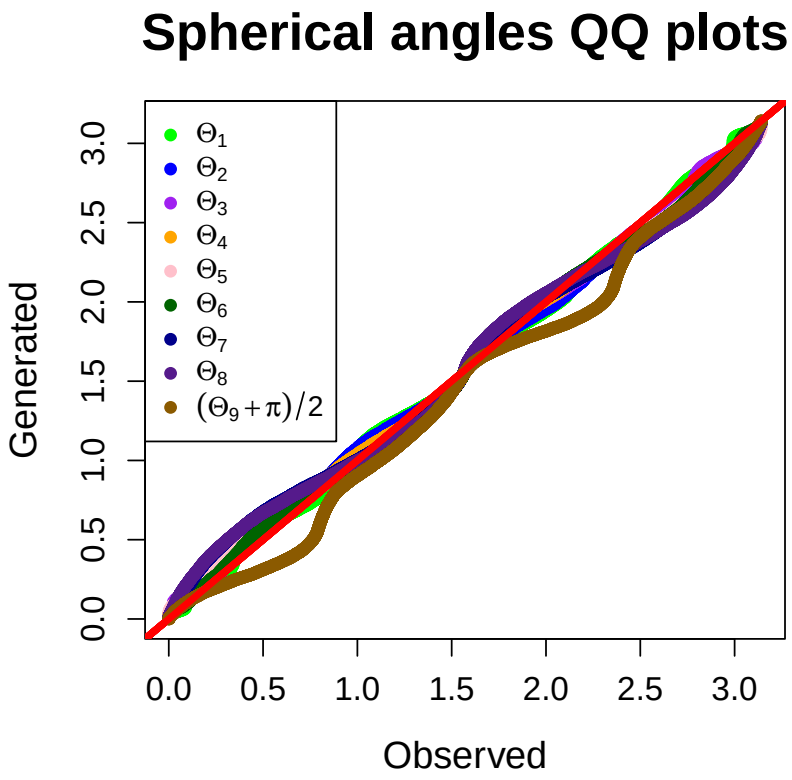
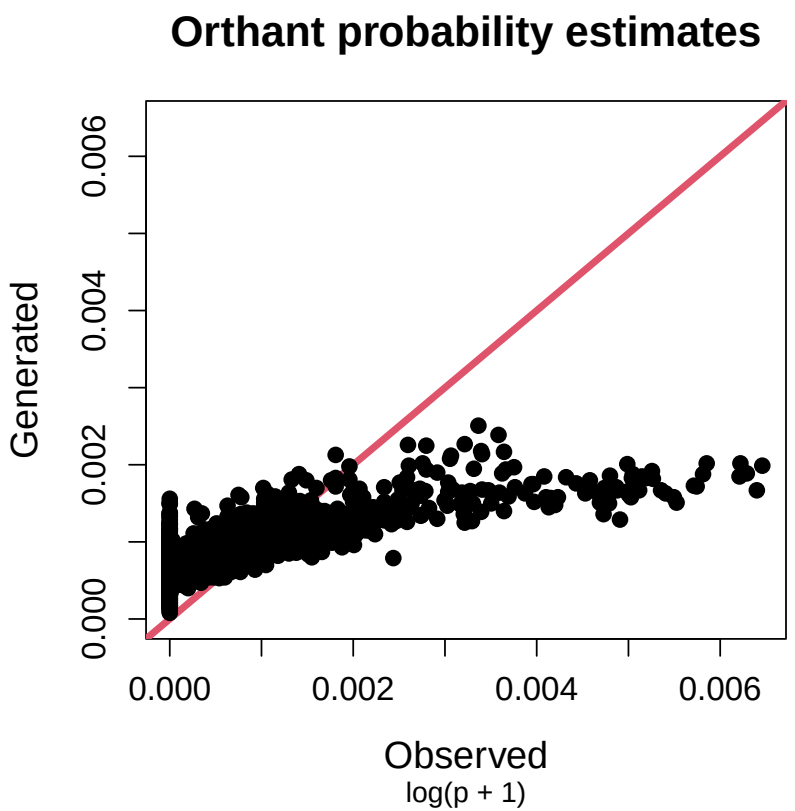
- The deep learning methods generally do well at recreating marginal *and* dependence structures.
  - There are some cases where the GAN doesn't do so well.
  - But there is no clear overall "winner".
- Marginal distributions do have an effect.
  - In some cases, we do well for Laplace margins but poorly for the equivalent double Pareto case.

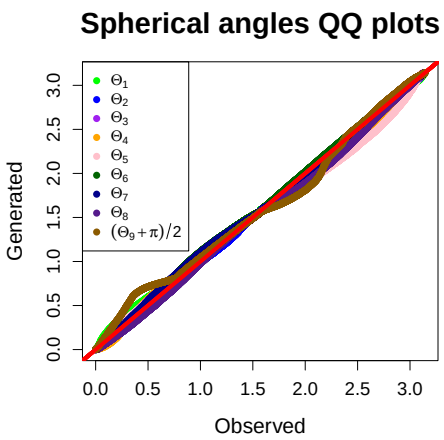
Simulation study - some example results

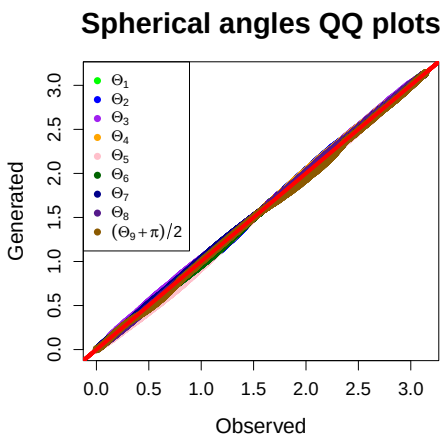
Sparse Gaussian copula, Double Pareto margins, $d = 10$ and $n = 100,000$

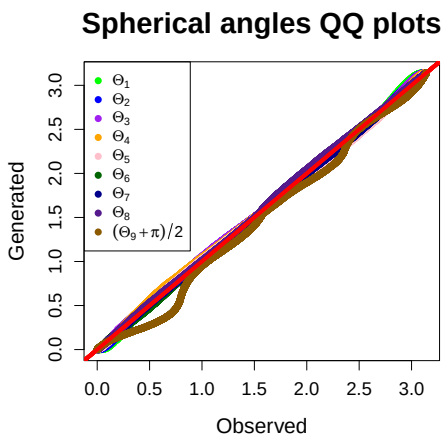


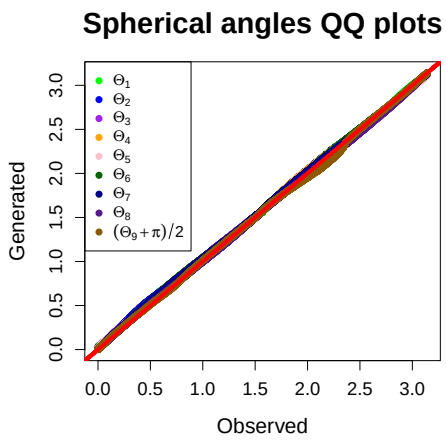


Simulation study - some example results









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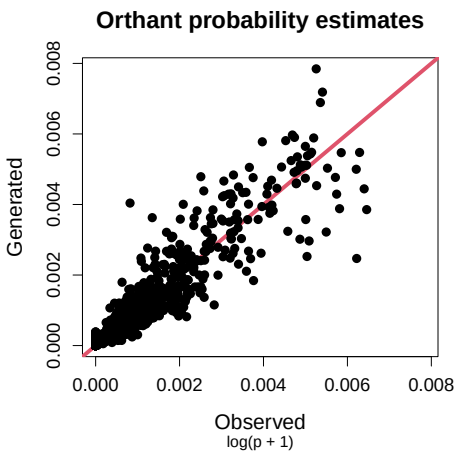
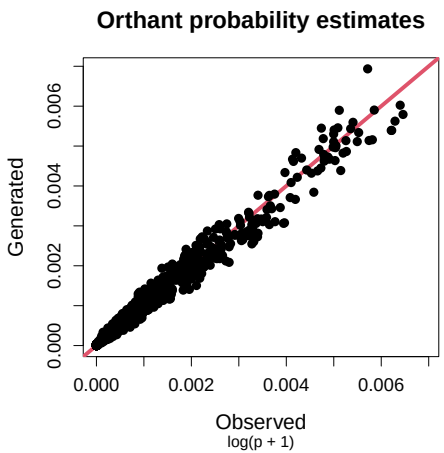
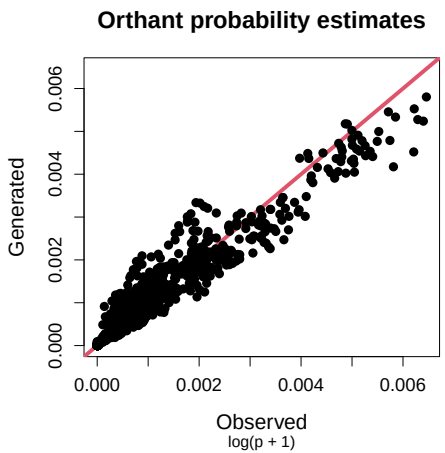
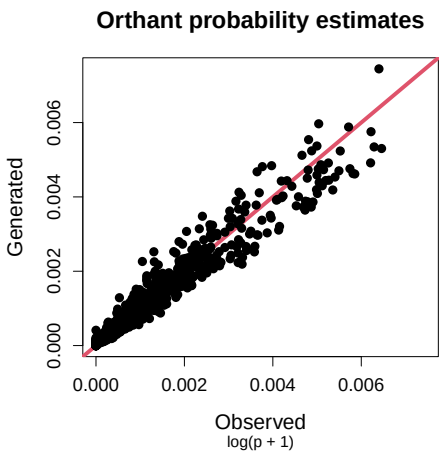






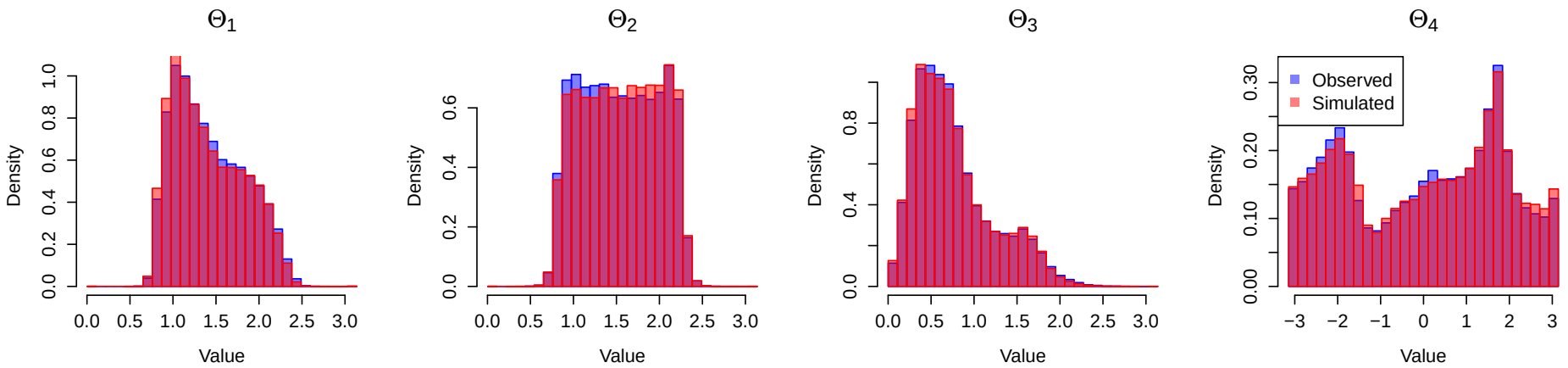


Metocean data example

Five-dimensional dataset consisting of:

- significant wave height,
- mean wave period,
- mean wave direction,
- hourly mean wind speed at 10 m above sea level,
- wind direction.

Metocean data - some example results



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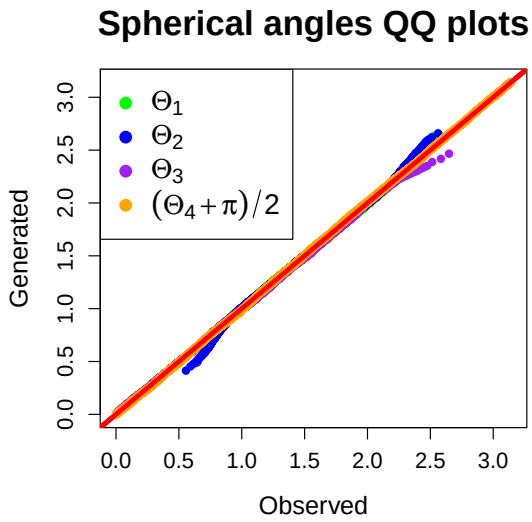

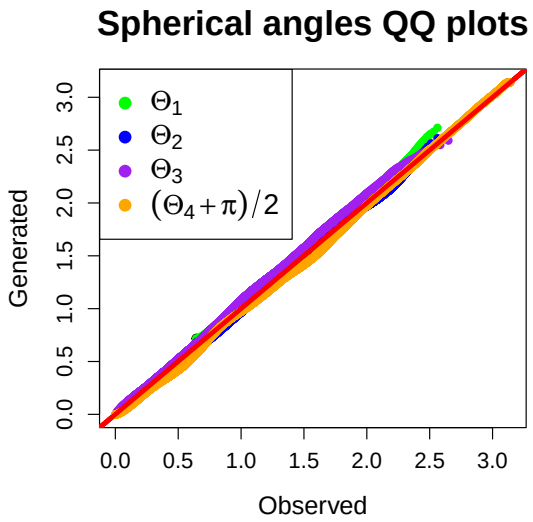

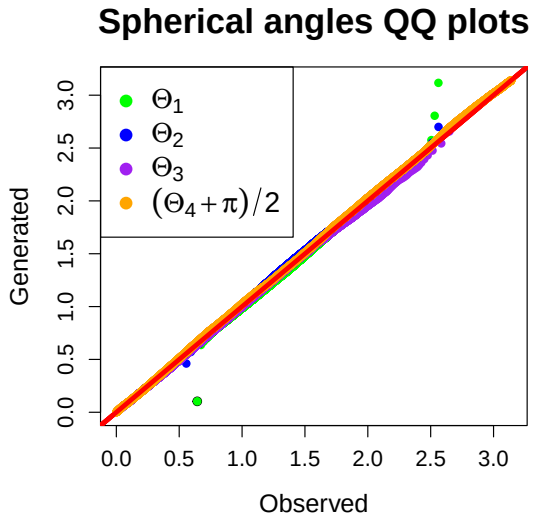

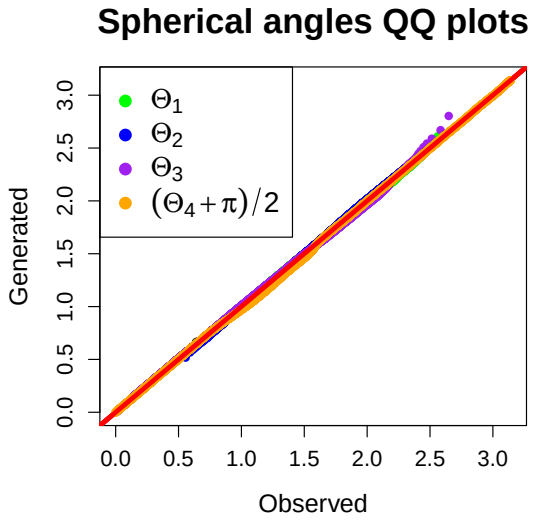

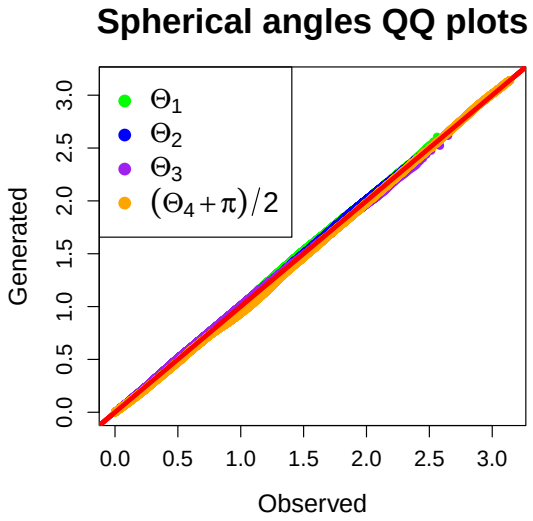

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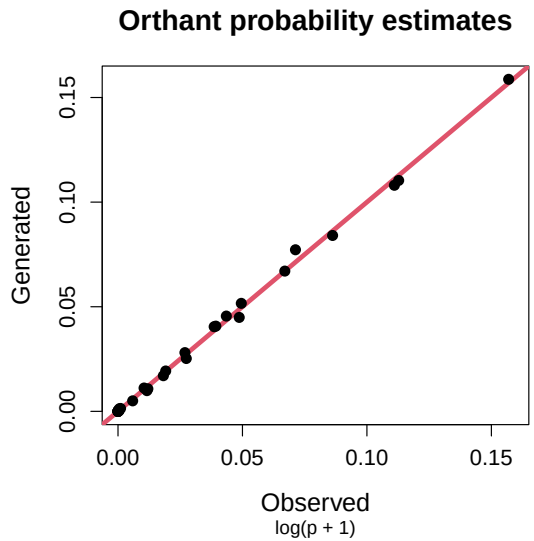

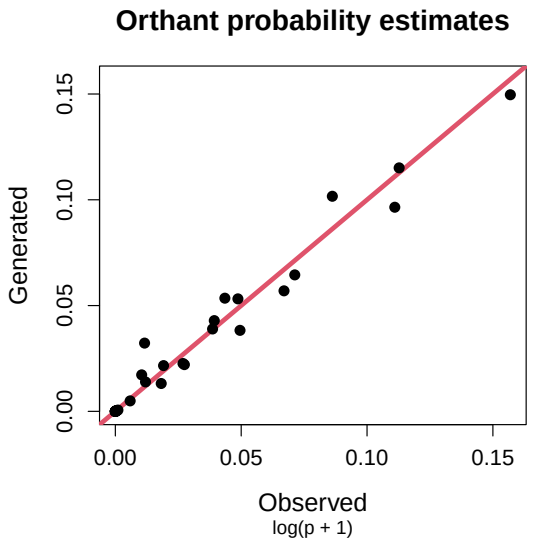

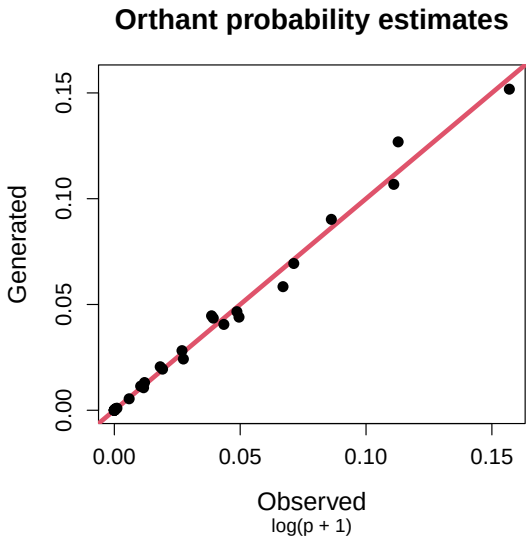

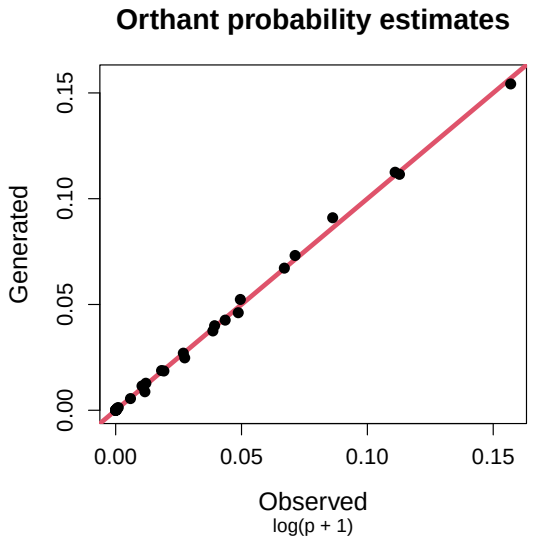

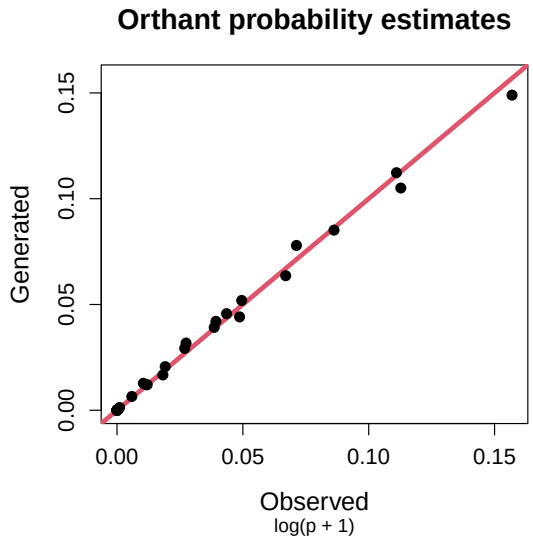

Overview

1. Background: Angular-radial representations in multivariate extremes
2. Multivariate angular simulation: The role of machine learning
3. Diagnostic techniques
4. Simulation study and a metocean application
5. Summary and outlook

Conclusions

- The flexibility and scalability offered by generative deep learning approaches make them suitable candidates for multivariate angular simulation.

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- The flexibility and scalability offered by generative deep learning approaches make them suitable candidates for multivariate angular simulation.
- The baseline parametric approach was generally good at capturing marginal distributions, but struggled with some complex dependence structures.
- No methods can be completely discounted, and there is no individual 'best' approach. So model checks and validation are key.

Avenues for further work

- Validation and model comparison is somewhat difficult.
  - It would be useful to have more metrics for evaluation in angular/spherical contexts.
  - Model rankings depend on the metric used.

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  - Theoretical developments or general guidelines related to this would be helpful.
- Improvements in computational efficiency could improve the feasibility of hyperparameter optimisation and uncertainty assessment.

Avenues for further work

- We focus on the L_2 norm, but some approaches may require different choices. Techniques for a more general setting may be useful.

- Our study is based on a specific set of machine learning approaches. This is a rapidly developing area, and others may be better suited to this angular simulation task.

Avenues for further work

- We focus on the L_2 norm, but some approaches may require different choices. Techniques for a more general setting may be useful.
- Our study is based on a specific set of machine learning approaches. This is a rapidly developing area, and others may be better suited to this angular simulation task.
- The Special Issue paper by Lhaut et al. (2026) also covers this topic.
  - Simulation of multivariate extremes: A Wasserstein–Aitchison GAN approach.
  - They apply a transformation to their angles to avoid restrictions to the sphere.

Thank you!

Based on the paper:

- Wessel, J. B., Murphy-Barltrop, C. J. R. and Simpson, E. S. (2026). Generative machine learning for multivariate angular simulation. *Extremes.*
- Accompanying code available from `https://github.com/callumbarltrop/DeGeMoH`

Other references:

- Durkan, C., Bekasov, A., Murray, I., and Papamakarios, G. (2019). Neural spline flows. *NeurIPS.*
- Goodfellow, I. J., et al. (2014). Generative adversarial nets. *NeurIPS.*
- Lipman, Y., Chen, R. T. Q., Ben-Hamu, H., Nickel, M., and Le, M. (2022). Flow matching for generative modeling. *ICLR.*
- Mackay, E. and Jonathan, P. (2023). Modelling multivariate extremes through angular-radial decomposition of the density function. *arXiv:2310.12711.*
- Papamakarios, G., Pavlakou, T., and Murray, I. (2017). Masked autoregressive flow for density estimation. *NeurIPS.*

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Evaluation strategies: Angular energy score

Given a (fitted) distribution F' on $\mathbb{S}^{d-1}$ and a fixed angle $\mathbf{v} \in \mathbb{S}^{d-1}$, the angular energy score is given by

$$\text{aES}(F', \mathbf{v}) = \mathbb{E}_{\mathbf{W} \sim F'}\{\alpha(\mathbf{W}, \mathbf{v})\} - \frac{1}{2}\mathbb{E}_{\mathbf{W}, \mathbf{W}' \sim F'}\left\{\alpha\left(\mathbf{W}, \mathbf{W}'\right)\right\}.$$

- $\alpha(\cdot, \cdot)$ is the angular distance between two points on $\mathbb{S}^{d-1}$;
- $\mathbf{W}$ and $\mathbf{W}'$ denote independent copies of the angular random variable with distribution F'.

Evaluation strategies: Angular energy score

- The aES is defined for model assessment at a fixed angle $\mathbf{v} \in \mathbb{S}^{d-1}$.
- We therefore calculate the expected aES over the 'true' angular distribution F.

- Closed-form expressions of F and F' may be unavailable, so Monte Carlo techniques are used for evaluation.